

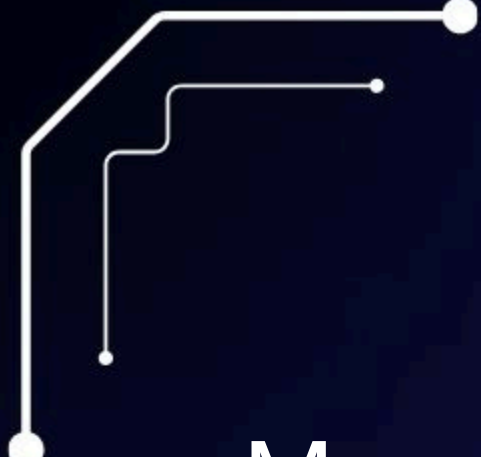
MOST AI AGENTS FAIL IN PRODUCTION

The problem usually is not the
model.

It is the architecture.

Here are three design patterns
teams use to build reliable AI
agents.





Many early agent prototypes follow a simple flow:

User → LLM → Response

That works for demos.

Real systems are different. Agents must interact with tools, handle multi step workflows, and verify results.

This is where agent design patterns become important.



01

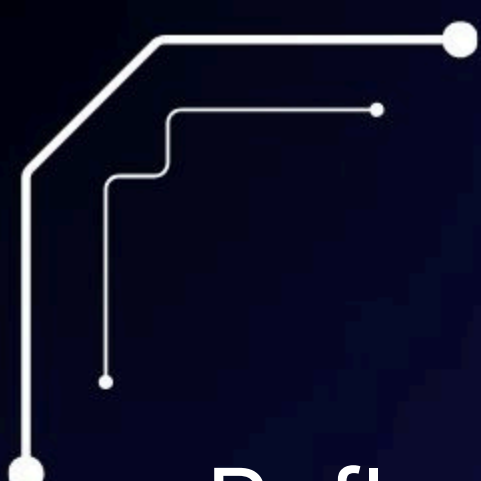
Reflection Pattern

The agent reviews its own output before finishing a task. It evaluates questions like:

Is the answer accurate? Did I miss anything important? Should I revise the result?

This extra step helps catch errors before they reach the user.





Reflection becomes especially useful
in multi step tasks.

For example:

An agent generating a report may
validate sources before finalizing the
output. Small corrections during the
process prevent larger failures later.



02

ReAct (Reason + Act) Pattern

Instead of generating a single response, the agent works in cycles.

- Reason about the task
- Use a tool or API
- Observe the result
- Adjust the next step

Modern agent frameworks already implement this loop. The real challenge is designing the right tools and workflows.



This pattern allows agents to interact with real systems:

- Databases
- CRMs
- Internal APIs

Reliability improves when agents move from “text generation” to “system interaction”





03

Multi Agent Systems Pattern

Complex workflows are often handled by multiple specialized agents.

Examples:

A research agent gathers information.

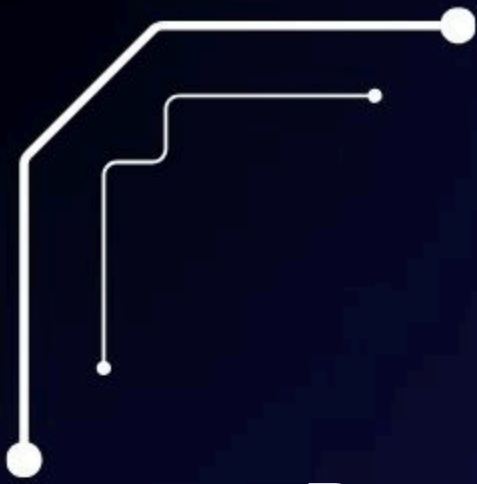
A writing agent drafts the content.

A review agent validates the output.

Each agent focuses on a specific role.

This improves reliability and scalability





Production AI agents behave more like systems than chatbots.

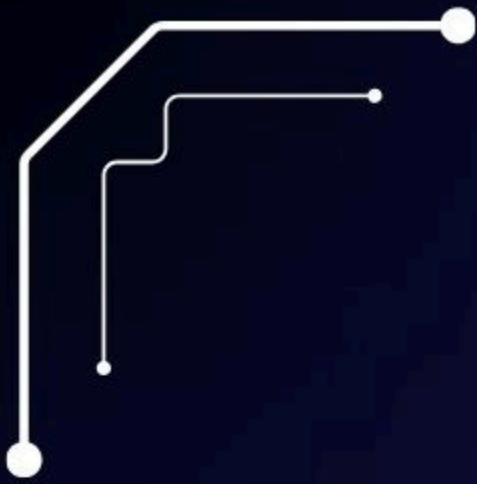
Many successful deployments combine these patterns:

Agents use tools through ReAct.

Reflection improves output quality.

Multiple agents collaborate on complex workflows.





AI agents become far more reliable when designed with the right architecture.

What design patterns have worked well for your AI systems?